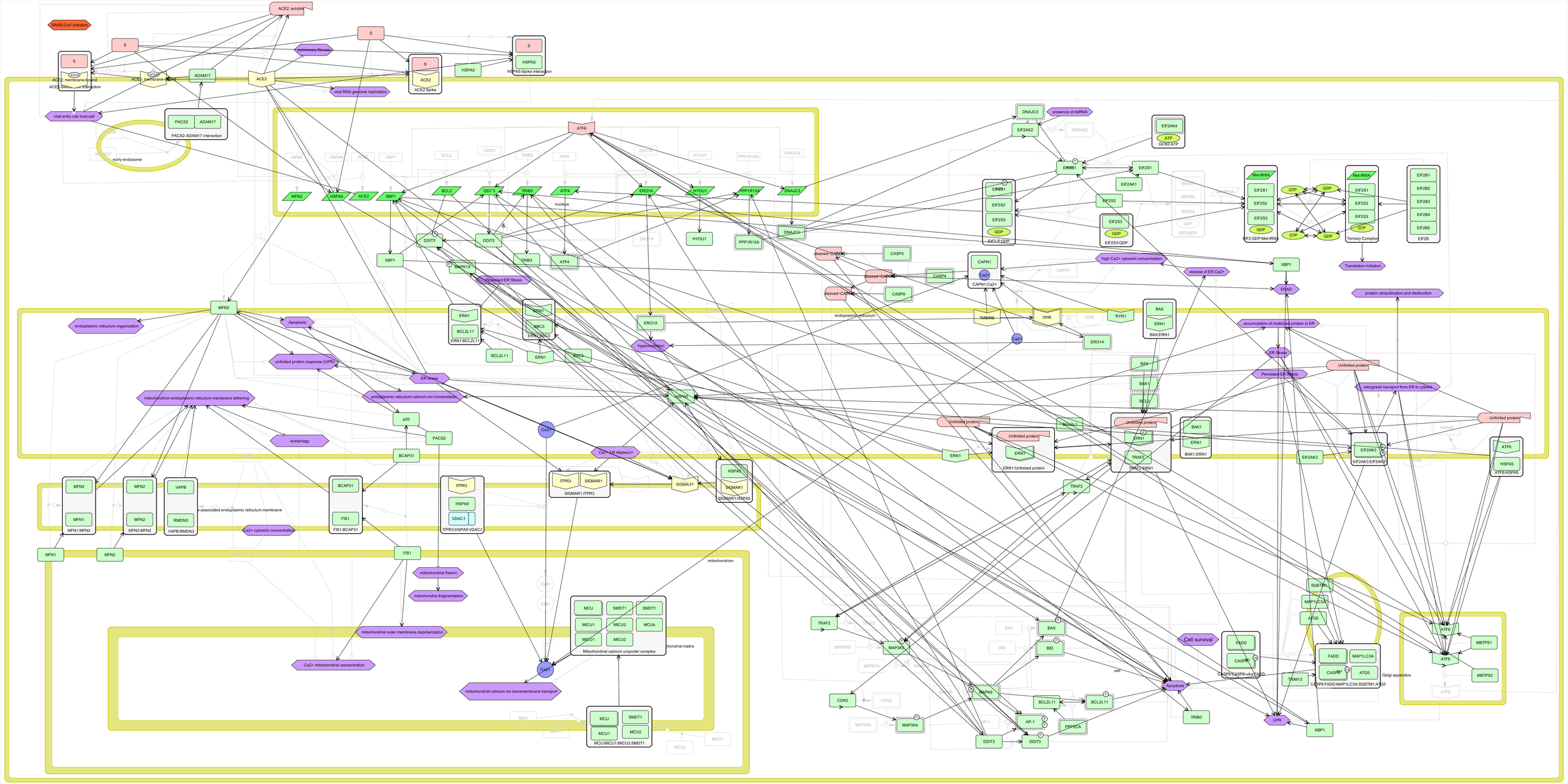

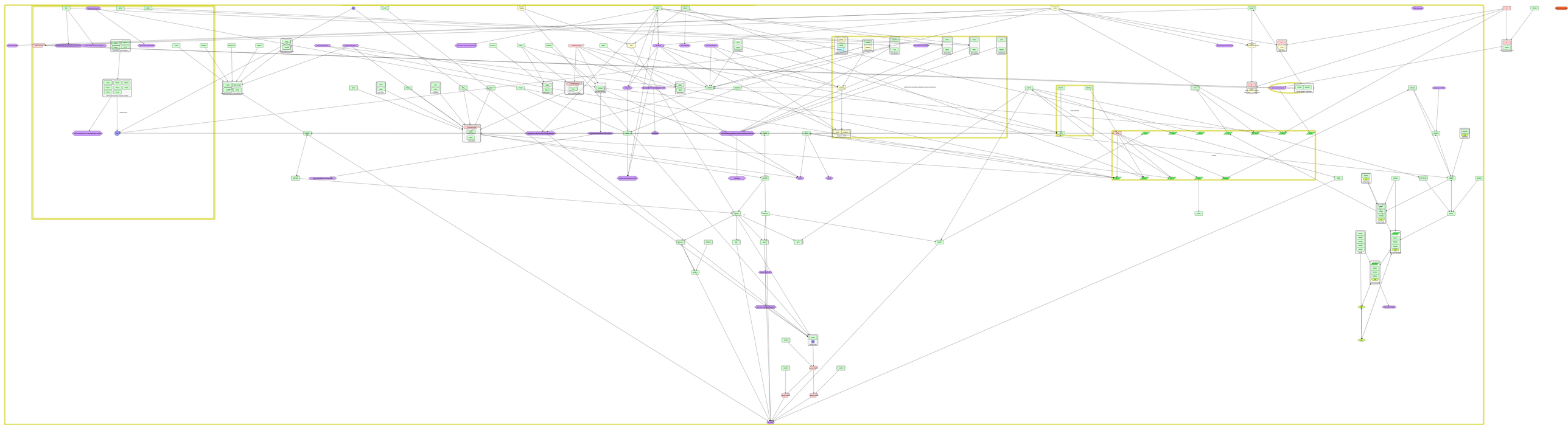

SARS-CoV infection
ACE2, soluble
S
S
S
S
S
HSPA5
HSPA5
pulmonary fibrosis
active
active
ACE2, membrane-bound
ACE2-SARS-CoV interaction
ACE2, membrane-bound
ADAM17
ACE2
ACE2
ACE2:Spike
HSPA5-Spike interaction
viral RNA genome replication
viral entry into host cell
PACS2
ADAM17
PACS2-ADAM17 interaction
PACS2
ADAM17
early endosome
ATF6
MFN2
HSPA5
ACE2
XBP1
BCL2
DDIT3
TRIB3
ATF4
ERO1A
HYOU1
PPP1R15A
DNAJC3
nucleus
&
DDIT3
DDIT3
XBP1
TRIB3
ATF4
MAPK14
P
Persistant ER Stress
HYOU1
PPP1R15A
DNAJC3
ERO1A
cleaved~CASP3
CASP3
cleaved~CASP4
CASP4
cleaved~CASP9
CASP9
CAPN1
Ca2+
CAPN1:Ca2+
CAPN1
DNAJC3
presence of dsRNA
EIF2AK2
EIF2AK2
EIF2AK4
ATP
GCN2:ATP
EIF2S1
P
EIF2S1
EIF2AK1
EIF2S2
EIF2S1
EIF2S2
EIF2S3
GDP
EIF2-P:GDP
EIF2S3
GDP
EIF2S3:GDP
EIF2S1
EIF2S2
EIF2S3
GDP
EIF2:GDP
Met-tRNA
Met-tRNA
EIF2S1
EIF2S2
EIF2S3
GDP
EIF2:GDP:Met-tRNA
GTP
GDP
GTP
GDP
Met-tRNA
EIF2S1
EIF2S2
EIF2S3
GTP
Ternary Complex
EIF2B1
EIF2B2
EIF2B3
EIF2B4
EIF2B5
EIF2B
Translation initiation
high Ca2+ cytosolic concentration
release of ER Ca2+
XBP1
ERAD
protein ubiquitination and destruction
MFN2
endoplasmic reticulum organization
Apoptosis
ERN1
BCL2L11
ERN1:BCL2L11
ERN1
BBC3
ERN1:BBC3
BCL2L11
ERN1
BBC3
ERO1A
hyperoxidation
endoplasmic reticulum
TMBIM6
ITPR
ITPR
RYR1
Ca2+
ERO1A
BAX
ERN1
BAX:ERN1
accumulation of misfolded protein in ER
ER Stress
Persistant ER Stress
Unfolded protein
retrograde transport from ER to cytosol
unfolded protein response (UPR)
ER stress
mitochondrion-endoplasmic reticulum membrane tethering
endoplasmic reticulum calcium ion homeostasis
HSPA5
p20
PACS2
autophagy
BCAP31
Ca2+
Ca2+ ER depletion
BAX
BAK1
BCL2
BCL2L1
Unfolded protein
Unfolded protein
ERN1
ERN1:Unfolded protein
Unfolded protein
P
ERN1
TRAF2
TRAF2:ERN1
BAK1
ERN1
BAK1:ERN1
ERN1
EIF2AK3
EIF2AK3
P
P
EIF2AK3:EIF2AK3
Unfolded protein
HSPA5
ATF6
HSPA5
ATF6:HSPA5
MFN2
MFN1
MFN1:MFN2
MFN2
MFN2
MFN2:MFN2
VAPB
RMDN3
VAPB:RMDN3
VAPB
mitochondria-associated endoplasmic reticulum membrane
Ca2+ cytosolic concentration
BCAP31
FIS1
FIS1:BCAP31
ITPR3
HSPA9
VDAC1
ITPR3:HSPA9:VDAC1
ITPR3
SIGMAR1
SIGMAR1:ITPR3
SIGMAR1
HSPA5
SIGMAR1
SIGMAR1:HSPA5
TRAF2
MFN1
MFN2
RMDN3
FIS1
mitochondrion
mitochondrial fission
mitochondria fragmentation
Ca2+
Ca2+
MCU
SMDT1
SMDT1
MICU1
MICU2
MCUb
MICU1
MICU2
Mitochondrial calcium uniporter complex
mitochondrial matrix
mitochondrial outer membrane depolarization
Ca2+ mitochondrial concentration
Ca2+
mitochondrial calcium ion transmembrane transport
MCU
SMDT1
MCU
MCU1
SMDT1
MCU2
MCU:MICU1:MICU2:SMDT1
MCU1
MCU2
TRAF2
TRAF2
MAP3K5
MAP3K5
P
MAPK14
MAPK14
P
BAX
BAX
P
BID
BID
P
Cell survival
FADD
CASP8
K63
Ub
CASP8:CASP8-ubq:FADD
SQSTM1
MAP1LC3A
ATG5
FADD
MAP1LC3A
CASP8
K63
Ub
ATG5
CASP8:FADD:MAP1LC3A:SQSTM1:ATG5
Golgi apparatus
TRIM13
ATF6
MBTPS1
ATF6
MBTPS2
ATF6
cell
MAPK8
MAPK8
P
BCL2L11
BCL2L11
P
Apoptosis
TRIB3
CDK5
?
CDK5
MAP3K4
MAP3K4
P
AP-1
AP-1
P
P
PPP2CA
DDIT3
DDIT3
P
UPR
XBP1

FIS1
mitochondrial fission
MFN2
MFN1
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MICU1
MICU2
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ATF6:HSPA5
ITPR3
HSPA9
VDAC1
ITPR3:HSPA9:VDAC1
HSPA5
SIGMAR1
SIGMAR1:HSPA5
BCAP31
FIS1
FIS1:BCAP31
Ca2+ cytosolic concentration
MFN2
MFN2
MFN2:MFN2
MFN2
MFN1
MFN1:MFN2
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RMDN3
VAPB:RMDN3
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MICU2
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FADD
MAP1LC3A
CASP8
ATG5
CASP8:FADD:MAP1LC3A:SQSTM1:ATG5
BAK1
BAK1
ERN1
BAK1:ERN1
BCL2L1
BAX
ERN1
BAX:ERN1
BAK
BCL2
PACS2
ERN1
BCL2L11
ERN1:BCL2L11
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TRAF2:ERN1
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p38
ER Stress
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XBP1
ITPR3
SIGMAR1
SIGMAR1:ITPR3
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ATF6
BCL2
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GDP
EIF2S3:GDP
EIF2S2
PPP1R15A
EIF2S1
EIF2AK1
MAPK8
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MAPK14
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EIF2S2
EIF2S3
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EIF2-P:GDP
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EIF2S2
EIF2S3
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BCL2L11
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EIF2S2
EIF2S3
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GTP
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GDP
CAPN1
Ca2+
CAPN1:Ca2+
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CASP3
cleaved~CASP4
CASP9
cleaved~CASP3
cleaved~CASP9
Apoptosis